MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 4 (Units 10, 11, 12 & 13): Tangents, Asymptotes, Curvature & Multivariable Calculus

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Module 4 Overview

Units Covered in Module 4:

- **Unit 10:** Tangent and Normal, Asymptotes, and Curvature / Radius of Curvature (ρ).
- **Unit 11:** Functions of Two Variables (Domain, Limit, Continuity, Partial & Total Derivatives).
- **Unit 12:** Homogeneous Functions and Euler's Theorem (1st & 2nd Order Proofs and Solved Exam Problems).
- **Unit 13:** Jacobian (J) and Hessian ($H(f)$) Determinants.

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	✓ **This file**
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

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1 Unit 10: Tangent and Normal, Asymptotes, and Curvature

1.1 Tangent and Normal Lines

For a curve $y = f(x)$ or $F(x, y) = 0$ at a point $P(x_1, y_1)$ with slope $m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)}$:

Equation of Tangent Line: $y - y_1 = m(x - x_1)$,

Equation of Normal Line: $y - y_1 = -\frac{1}{m}(x - x_1)$ [Slope of normal $= -1/m$].

PUST EECE 2020–21 Semester Exam — Question 4(c) [4.5 Marks]
Question: Find the tangent and normal to the curve $x^2 + xy + y^2 = 7$ at the point $(1, 2)$.

Solution (Exam Style): Given curve:

$$x^2 + xy + y^2 = 7$$

Differentiating both sides implicitly with respect to x:

$$\begin{aligned} 2x + \left(x\frac{dy}{dx} + y\right) + 2y\frac{dy}{dx} &= 0 \\ \implies (x + 2y)\frac{dy}{dx} &= -(2x + y) \\ \implies \frac{dy}{dx} &= -\frac{2x + y}{x + 2y} \end{aligned}$$

At the point $(1, 2)$:

Slope of tangent, $m = \left(\frac{dy}{dx}\right)_{(1,2)} = -\frac{2(1) + 2}{1 + 2(2)} = -\frac{4}{5}$.

Equation of the Tangent at $(1, 2)$:

$$\begin{aligned} y - 2 &= -\frac{4}{5}(x - 1) \\ \implies 5y - 10 &= -4x + 4 \\ \implies 4x + 5y - 14 &= 0. \end{aligned}$$

Equation of the Normal at $(1, 2)$:

$$\begin{aligned} y - 2 &= \frac{5}{4}(x - 1) \quad [\text{Slope of normal} = -\tfrac{1}{m} = \tfrac{5}{4}] \\ \implies 4y - 8 &= 5x - 5 \\ \implies 5x - 4y + 3 &= 0. \end{aligned}$$

1.2 Asymptotes of Algebraic Curves

Definition (Asymptote): A straight line is called an asymptote to a curve if the perpendicular distance from a point $P(x, y)$ on the curve to the line approaches zero as P moves to infinity along a branch of the curve.

Working Rule for Oblique Asymptotes ($y = mx + c$) of an n-th Degree Curve $\phi_n(x, y) + \phi_{n-1}(x, y) + \cdots = 0$:

1. Put $x = 1$ and $y = m$ in the homogeneous parts of degree n and $n - 1$ to obtain $\phi_n(m)$ and $\phi_{n-1}(m)$.
2. Solve $\phi_n(m) = 0$ to find the slopes m.
3. For each non-repeated root m, find c using:

$$c = -\frac{\phi_{n-1}(m)}{\phi_n'(m)}$$

and substitute (m, c) into $y = mx + c$.

PUST EECE 2022–23 Semester Exam — Question 3(b) [5 Marks]
Question: Define asymptotes. Find the asymptotes of the curve $x^3 + y^3 - 3axy = 0$.

Solution (Exam Style):

1. Since the coefficients of the highest powers x^3 and y^3 are constants ($1 \neq 0$), there are no asymptotes parallel to the x-axis or y-axis.
2. Putting $x = 1, y = m$ in the homogeneous parts of degree 3 and degree 2:

$$\phi_3(m) = 1 + m^3 \implies \phi_3'(m) = 3m^2$$
$$\phi_2(m) = -3am$$

3. Finding slope m:

$$\begin{aligned}\phi_3(m) &= 0\\ \implies m^3 + 1 &= 0\\ \implies (m+1)(m^2 - m + 1) &= 0\\ \implies m &= -1 \quad \text{[Only real root]}\end{aligned}$$

4. Finding intercept c for $m = -1$:

$$c = -\frac{\phi_2(m)}{\phi_3'(m)} = -\frac{-3am}{3m^2} = \frac{a}{m}$$
$$\implies c = \frac{a}{-1} = -a$$

5. Substituting $m = -1, c = -a$ into $y = mx + c$:

$$\begin{aligned}y &= -x - a\\ \implies x + y + a &= 0.\end{aligned}$$

PUST EECE 2020–21 Semester Exam — Question 3(c) [4.5 Marks]
Question: Find the asymptotes of $y^3 - x^2y - 2xy^2 + 2x^3 - 7xy + 3y^2 + 2x^2 + 2x + 2y + 1 = 0$.

Solution (Exam Style): Putting $x = 1, y = m$ in the third- and second-degree homogeneous terms:

$$\phi_3(m) = m^3 - 2m^2 - m + 2 = (m-1)(m+1)(m-2),$$
$$\phi_3'(m) = 3m^2 - 4m - 1,$$
$$\phi_2(m) = 3m^2 - 7m + 2.$$

Setting $\phi_3(m) = 0$ gives $m = 1,\ -1,\ 2$. Using $c = -\dfrac{\phi_2(m)}{\phi_3'(m)} = -\dfrac{3m^2 - 7m + 2}{3m^2 - 4m - 1}$:

$$\text{For } m = 1: \quad c = -\frac{3 - 7 + 2}{3 - 4 - 1} = -1 \implies y = x - 1 \quad (x - y - 1 = 0),$$

$$\text{For } m = -1: \quad c = -\frac{3 + 7 + 2}{3 + 4 - 1} = -2 \implies y = -x - 2 \quad (x + y + 2 = 0),$$

$$\text{For } m = 2: \quad c = -\frac{12 - 14 + 2}{12 - 8 - 1} = 0 \implies y = 2x \quad (2x - y = 0).$$

1.3 Curvature (κ) and Radius of Curvature ($\rho = 1/\kappa$)

Formulas for Radius of Curvature (ρ):

Cartesian Form: $\rho = \dfrac{\left(1 + y_1^2\right)^{3/2}}{|y_2|}$, Parametric Form: $\rho = \dfrac{\left(x'^2 + y'^2\right)^{3/2}}{|x'y'' - y'x''|}$, Newton's Formula at Origin: $\rho = \lim\limits_{x \to 0} \left|\dfrac{x^2}{2y}\right|$

PUST EECE 2022–23 Semester Exam — Question 3(c) [4 Marks]

Question: Find the radius of curvature at the point $\left(\frac{a}{4}, \frac{a}{4}\right)$ of the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$.

Solution (Exam Style):

Given, $x^{1/2} + y^{1/2} = a^{1/2}$

$$\implies \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\, y_1 = 0 \quad \text{[Differentiating w.r.t. } x\text{]}$$

$$\implies y_1 = -\frac{\sqrt{y}}{\sqrt{x}} = -y^{1/2}x^{-1/2}$$

$$\text{At } \left(\frac{a}{4}, \frac{a}{4}\right): \quad y_1 = -\frac{\sqrt{a/4}}{\sqrt{a/4}} = -1.$$

Differentiating $y_1 = -y^{1/2}x^{-1/2}$ again with respect to x:

$$y_2 = -\left[\frac{1}{2}y^{-1/2}y_1x^{-1/2} + y^{1/2}\left(-\frac{1}{2}x^{-3/2}\right)\right]$$

$$= -\frac{1}{2\sqrt{xy}}\left(-\frac{\sqrt{y}}{\sqrt{x}}\right) + \frac{\sqrt{y}}{2x\sqrt{x}}$$

$$= \frac{\sqrt{x} + \sqrt{y}}{2x^{3/2}} = \frac{\sqrt{a}}{2x^{3/2}} \quad [\because \sqrt{x} + \sqrt{y} = \sqrt{a}]$$

$$\text{At } x = \frac{a}{4}: \quad y_2 = \frac{\sqrt{a}}{2(a/4)^{3/2}} = \frac{\sqrt{a}}{2\left(\frac{a\sqrt{a}}{8}\right)} = \frac{4}{a}.$$

Substituting $y_1 = -1$ and $y_2 = \dfrac{4}{a}$ into the radius of curvature formula:

$$\rho = \frac{\left(1 + y_1^2\right)^{3/2}}{|y_2|}$$

$$= \frac{\left(1 + (-1)^2\right)^{3/2}}{4/a} = \frac{2\sqrt{2}\,a}{4} = \frac{a}{\sqrt{2}}.$$

PUST EECE 2020–21 Exam Q4(b) [4.5 Marks] & 2021–22 Exam Q3(b) [5 Marks]

Problem A (2020–21 Q4b): Show that the radius of curvature at (x, y) on $x^{2/3} + y^{2/3} = a^{2/3}$ is $3(axy)^{1/3}$.

Solution (Exam Style):

$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$\implies \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y_1 = 0 \implies y_1 = -\frac{y^{1/3}}{x^{1/3}}$$

$$\implies 1 + y_1^2 = 1 + \frac{y^{2/3}}{x^{2/3}} = \frac{x^{2/3} + y^{2/3}}{x^{2/3}} = \frac{a^{2/3}}{x^{2/3}}$$

$$\implies y_2 = -\frac{d}{dx}\left(y^{1/3}x^{-1/3}\right) = \frac{a^{2/3}}{3x^{4/3}y^{1/3}}$$

$$\therefore \rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{a/x}{\dfrac{a^{2/3}}{3x^{4/3}y^{1/3}}} = 3(axy)^{1/3}. \quad \blacksquare$$

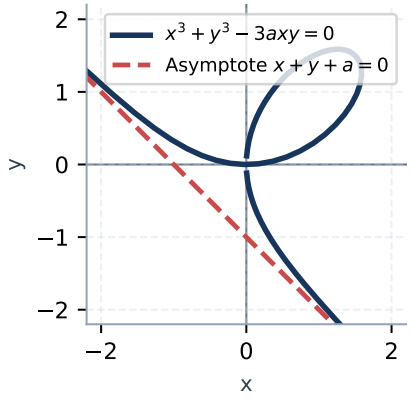
Problem B (2021–22 Q3b): Find the radius of curvature at $(a, 0)$ on $xy^2 = a^3 - x^3$.

Solution (Exam Style): Shifting origin to $(a, 0)$ by putting $x = a + X$, $y = Y$:

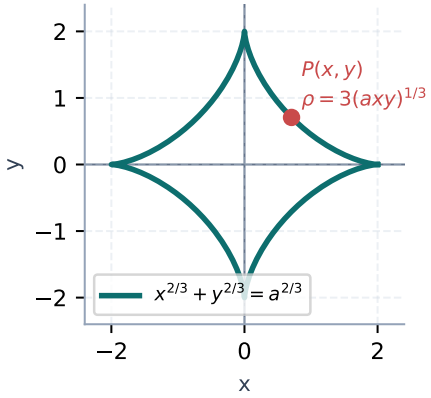
$$(a + X)Y^2 = a^3 - (a + X)^3 = -3a^2X - 3aX^2 - X^3$$

$$\implies \rho = \lim_{Y\to 0}\left|\frac{Y^2}{2X}\right| = \lim_{X\to 0}\left|\frac{-3a^2 - 3aX - X^2}{2(a + X)}\right| = \frac{3a}{2} \quad \text{[By Newton's curvature formula].}$$

PUST 22-23 Q3(b): Folium of Descartes
Oblique Asymptote $x + y + a = 0$



PUST 20-21 Q4(b): Astroid Curve
Radius of Curvature $\rho = 3(axy)^{1/3}$



2 Unit 11: Functions of Two Variables, Partial & Total Derivatives

2.1 Domain, Limit, and Continuity of $f(x, y)$ (Class Lecture)

1. **Class Domain Examples:**

$$f(x, y) = \sqrt{y - x^2} \implies y - x^2 \geq 0 \implies D = \{(x, y) \in \mathbb{R}^2 : y \geq x^2\},$$

$$f(x, y) = \frac{1}{xy} \implies xy \neq 0 \implies D = \{(x, y) \in \mathbb{R}^2 : x \neq 0,\ y \neq 0\}.$$

2. **Class Limit Example:** Show that $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Along the path $y = mx$: $\quad L = \lim_{x\to 0} \dfrac{x^2 - m^2x^2}{x^2 + m^2x^2}$

$$= \frac{1-m^2}{1+m^2} \quad \text{[Depends on slope } m\text{, so limit does not exist]}.$$

2.2 Partial Derivatives, Implicit Partial Differentiation & Total Derivatives

- **Class Example:** For $f(x,y) = 2x^3y^2 + 2y + 4x$:

$$f_x = 6x^2y^2 + 4, \quad f_y = 4x^3y + 2, \quad f_{xx} = 12xy^2, \quad f_{yy} = 4x^3, \quad f_{xy} = f_{yx} = 12x^2y.$$

- **Implicit Partial Derivatives ($F(x,y,z) = 0$):** $\dfrac{\partial z}{\partial x} = -\dfrac{F_x}{F_z}, \qquad \dfrac{\partial z}{\partial y} = -\dfrac{F_y}{F_z}.$
- **Total Differential & Total Derivative:**

$$dz = \frac{\partial z}{\partial x}\,dx + \frac{\partial z}{\partial y}\,dy, \qquad \frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

3 Unit 12: Homogeneous Functions and Euler's Theorem

3.1 Statement and Proof of Euler's Theorem (PUST EECE 2020–21 Q3a)

Definition (Homogeneous Function): A function $u = f(x,y)$ is homogeneous of degree n if $f(tx,ty) = t^n f(x,y)$, or equivalently $u = x^n\phi\left(\dfrac{y}{x}\right)$.

> **Euler's Theorem on Homogeneous Functions (Statement & Proof) [5 Marks]**
> **Statement:** If $u = f(x,y)$ is a homogeneous function of x and y of degree n, then:
>
> (i) $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu,$ (ii) $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x\partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = n(n-1)u.$

Proof (Exam Style): Since u is homogeneous of degree n, let:

$$u = x^n\phi\left(\frac{y}{x}\right)$$

Differentiating partially with respect to x and y:

$$\frac{\partial u}{\partial x} = nx^{n-1}\phi\left(\frac{y}{x}\right) + x^n\phi'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) = nx^{n-1}\phi\left(\frac{y}{x}\right) - x^{n-2}y\phi'\left(\frac{y}{x}\right)$$
$$\frac{\partial u}{\partial y} = x^n\phi'\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) = x^{n-1}\phi'\left(\frac{y}{x}\right)$$

Multiplying by x and y respectively and adding:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nx^n\phi\left(\frac{y}{x}\right) - x^{n-1}y\phi'\left(\frac{y}{x}\right) + x^{n-1}y\phi'\left(\frac{y}{x}\right)$$
$$= nx^n\phi\left(\frac{y}{x}\right) = nu. \quad \blacksquare$$

3.2 Solved Class Lecture & PUST Past Exam Problems on Euler's Theorem

> **Class Lecture & PUST EECE 2021–22 Exam Q3(a) [4.5 Marks]**
> **Question:** If $u = \sin^{-1}\left(\dfrac{x^2+y^2}{x+y}\right)$, show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \tan u$.

Solution (Exam Style):

Given, $u = \sin^{-1}\left(\dfrac{x^2+y^2}{x+y}\right)$

$$\implies z = \sin u = \frac{x^2+y^2}{x+y} = x^1 \frac{1+(y/x)^2}{1+(y/x)} \quad \text{[Homogeneous of degree } n=1\text{]}$$

Applying Euler's Theorem to $z = \sin u$:

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 1 \cdot z$$
$$\implies x\frac{\partial}{\partial x}(\sin u) + y\frac{\partial}{\partial y}(\sin u) = \sin u$$
$$\implies x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = \sin u$$
$$\implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{\sin u}{\cos u} = \tan u \quad \text{[Dividing by } \cos u\text{]}. \ \blacksquare$$

PUST EECE 2022–23 Exam Q3(a) [5 Marks] & 2020–21 Exam Q3(b) [4.5 Marks]

Problem A (2022–23 Q3a): If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.

Solution (Exam Style):

Let $z = \tan u = \frac{x^3+y^3}{x-y} = x^2\frac{1+(y/x)^3}{1-(y/x)}$ [Homogeneous of degree $n = 3-1 = 2$]

$$\implies x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z \quad \text{[By Euler's Theorem]}$$
$$\implies x\sec^2 u\frac{\partial u}{\partial x} + y\sec^2 u\frac{\partial u}{\partial y} = 2\tan u$$
$$\implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{2\tan u}{\sec^2 u} = 2\sin u\cos u = \sin 2u. \ \blacksquare$$

Problem B (2020–21 Q3b): If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + \frac{1}{2}\cot u = 0$.

Solution (Exam Style):

Let $z = \cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}} = x^{1/2}\frac{1+(y/x)}{1+(y/x)^{1/2}}$ [Homogeneous of degree $n = 1-\frac{1}{2} = \frac{1}{2}$]

$$\implies x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{1}{2}z \quad \text{[By Euler's Theorem]}$$
$$\implies -x\sin u\frac{\partial u}{\partial x} - y\sin u\frac{\partial u}{\partial y} = \frac{1}{2}\cos u$$
$$\implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + \frac{1}{2}\cot u = 0 \quad \text{[Dividing by } -\sin u\text{]}. \ \blacksquare$$

4 Unit 13: Jacobian and Hessian Determinants

4.1 The Jacobian (J) and Its Properties (Class Lecture)

Definition (Jacobian): For $u = u(x,y)$ and $v = v(x,y)$:

$$J\left(\frac{u,v}{x,y}\right) = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x}.$$

Key Properties: (1) $J \cdot J' = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$; (2) Chain Rule: $\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)}$.

Problem 13.1 (Class Lecture): Find $\dfrac{\partial(x,y)}{\partial(r,\theta)}$ if $x = r\cos\theta$, $y = r\sin\theta$.

Solution (Exam Style):

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$
$$= r\cos^2\theta + r\sin^2\theta = r(\cos^2\theta + \sin^2\theta) = r.$$

PUST EECE 2021–22 Semester Exam — Question 4(c) [4.5 Marks]

Question: If $u = x^2 - 2y^2$ and $v = 2x^2 - y^2$, where $x = r\cos\theta$ and $y = r\sin\theta$, show that $\dfrac{\partial(u,v)}{\partial(r,\theta)} = 6r^3\sin 2\theta$.

Solution (Exam Style):

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 2x & -4y \\ 4x & -2y \end{vmatrix} = -4xy + 16xy = 12xy$$
$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$
$$\implies \frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)} \quad \text{[By Jacobian Chain Rule]}$$
$$= (12xy)\cdot r$$
$$= 12(r\cos\theta)(r\sin\theta)r \quad \text{[Substituting } x = r\cos\theta,\ y = r\sin\theta\text{]}$$
$$= 6r^3(2\sin\theta\cos\theta) = 6r^3\sin 2\theta. \quad \blacksquare$$

4.2 The Hessian Determinant ($H(f)$) (Class Lecture)

Definition (Hessian): For a twice-differentiable function $f(x,y)$, the Hessian determinant is:

$$H(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2.$$

At a stationary point (a,b) where $f_x(a,b) = f_y(a,b) = 0$:

- $H(a,b) > 0$ and $f_{xx}(a,b) < 0 \implies$ Local Maximum.
- $H(a,b) > 0$ and $f_{xx}(a,b) > 0 \implies$ Local Minimum.
- $H(a,b) < 0 \implies$ Saddle Point.

4.3 Past-Question Map for Module 4

Session	Questions Asked from Module 4 (Units 10–13)	Marks
2022–23	Q3(a) Euler's theorem for $u = \tan^{-1}\frac{x^3+y^3}{x-y}$ · Q3(b) Asymptotes of $x^3+y^3-3axy=0$ · Q3(c) Curvature of $\sqrt{x}+\sqrt{y}=\sqrt{a}$	$5+5+4$
2021–22	Q3(a) Euler's theorem for $u = \sin^{-1}\frac{x^2+y^2}{x+y}$ · Q3(b) Curvature at $(a,0)$ on $xy^2 = a^3 - x^3$ · Q4(c) Jacobian $\frac{\partial(u,v)}{\partial(r,\theta)} = 6r^3\sin 2\theta$	$4.5+5+4.5$
2020–21	Q3(a) State & prove Euler's theorem · Q3(b) $\cos^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$ · Q3(c) Asymptotes · Q4(b) Curvature of $x^{2/3}+y^{2/3}=a^{2/3}$ · Q4(c) Tangent/Normal	$5+4.5+4.5+4.5+4.5$

Quick Revision — Module 4 (Units 10–13)

- **Asymptotes & Curvature:** $\phi_n(m)=0,\ c=-\frac{\phi_{n-1}(m)}{\phi_n'(m)},\ y=mx+c;\quad \rho=\frac{(1+y_1^2)^{3/2}}{|y_2|};$ Origin Newton formula: $\rho=\lim\left|\frac{x^2}{2y}\right|$ or $\lim\left|\frac{y^2}{2x}\right|$.
- **Euler's Theorem:** If $z=f(u)$ is homogeneous of degree n: $xu_x+yu_y=\frac{nf(u)}{f'(u)}$.
- **Jacobian & Hessian:** $\frac{\partial(u,v)}{\partial(r,\theta)}=\frac{\partial(u,v)}{\partial(x,y)}\cdot\frac{\partial(x,y)}{\partial(r,\theta)};\quad$ Polar $\frac{\partial(x,y)}{\partial(r,\theta)}=r;\quad H=f_{xx}f_{yy}-(f_{xy})^2.$

Common Mistakes to Avoid in the Exam

- Applying Euler's theorem directly to $u=\sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$: u is not homogeneous! Always rewrite as $z=\sin u=\frac{x^2+y^2}{x+y}$ (degree $n=1$) first.
- Forgetting the minus sign in the oblique asymptote intercept formula $c=-\frac{\phi_{n-1}(m)}{\phi_n'(m)}$.